

Keith Hume Fraser Reserve Redevelopment

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95 Swansea Rd, Montrose — Project Summary & Proposal

Prepared for Yarra Ranges Council and Montrose Cricket Club — master plan rev 9.30

The proposal in brief

Keith Hume Fraser Reserve is an 11.95 ha reserve carrying two tired, poorly drained football pitches laid out in a way that wastes the ground available. The pressure on them is not hypothetical: Lilydale Montrose United fields approximately 430 players in 2026, has grown 5–10% a year, and already hires an additional ground on Quarry Rd, Lilydale to fit preseason and season fixtures — demand the reserve cannot currently hold. This proposal redevelops the playing precinct into **four football pitches and two full-size community cricket grounds** on the same footprint, served by a new central pavilion, without touching the bushland that gives the reserve its character.

The layout places all four pitches on a single 310° alignment on the Swansea Rd side of the watercourse, in two paired precincts: the south ground (Pitch 1 at 88 × 60 m and Pitch 2 at 105 × 68 m) and the north ground (Pitch 3 at 100 × 68 m and Pitch 4 at 75 × 56.4 m). Between each pair sits a concrete cricket wicket with glued carpet, protected through winter by mobile technical-area shelters that roll over the wicket for the football season. Each pair is enclosed by a single fenced, laser-flat grass field, so that in summer each precinct converts to one continuous cricket ground: **C1 at a 110 m boundary and C2 at 100 m — genuine senior community sizes, fixed in the design.**

A new pavilion sits between Pitches 2 and 3 on a podium about 3 m above playing level — a 70 × 25 m enclosed building containing six player change rooms, two referee rooms, a first-aid room doubling as the uniform shop, and a canteen with serving windows to both Pitch 2 and Pitch 3 plus an internal window to the main club room. At its northern end the structure carries a **20 × 20 m elevated open deck for outdoor table viewing across Pitches 2 and 3**, and above the building a smaller rooftop viewing area sights **all four pitches** — the direct Montrose expression of Aspendale's celebrated elevated viewing deck. Every part of the plan is dimensioned in MGA55 coordinates and has been checked against pitch run-off standards, cricket boundary-buffer guidance, waterway offsets, and the parcel boundary.

The precedent: Aspendale Gardens Reserve

This is not a speculative concept — it is the model the City of Kingston has already delivered at Aspendale Gardens, and the parallel is close to exact. That project produced a makeover of four football pitches and two cricket ovals plus training nets and twelve modern lighting poles, anchored by a contemporary single-storey pavilion housing four changerooms with amenities, a large kitchen and bar, a generous multi-purpose social space, and referee and first-aid rooms. Notably for our site, Aspendale's flood-level constraint was answered by raising the building onto an elevated amphitheatre-style platform with viewing across all pitches — precisely the role of our 3 m pavilion podium beside the Montrose drain. Aspendale was also delivered in stages — the sporting fields and car parking first, the pavilion second — and this proposal adopts the same discipline across three stages: **Stage 1** redevelops the lower oval into Pitches 3 and 4 with their cricket wicket and north-ground lighting, while the existing club rooms keep operating; **Stage 2** builds the new pavilion on its podium between the grounds; **Stage 3** removes the old rooms, builds Pitches 1 and 2 with the south cricket ground, and delivers the new ~220-space car park off the tennis-club entrance. Montrose can point to Aspendale as proof that a council-delivered project of exactly this shape works, is fundable in stages, and is loved by its clubs.

Reference: <https://shape.com.au/projects/aspendale-gardens-reserve-pavilion-city-of-kingston/>

Delivery and funding — how Aspendale got built

The numbers behind the precedent are as persuasive as the design. Kingston delivered the full Aspendale Gardens redevelopment in partnership with the Victorian Government, which contributed \$4.5 million alongside council's \$4.6 million — close to a 50/50 split on a roughly \$9.1 million ground transformation. The Stage 2 pavilion was a \$3.6 million project, again jointly funded by the state and council. The same state programs that funded Aspendale exist for Yarra Ranges: this proposal is not asking council to carry the project alone, it is asking council to sponsor a proven, co-funded model.

Delivery method matters as much as money on a constrained residential frontage. The Aspendale pavilion was built as a modular design-and-construct — fabricated off-site and craned into place — cutting construction noise for neighbours and keeping disruption to the operating clubs to a minimum. Given Swansea Rd's residential outlook and the intent that sport continues through both stages, this proposal adopts the same modular approach for the KHFR pavilion from the outset.

Kingston's advocacy case also travels directly to Montrose: purpose-built female-friendly change facilities for the fastest-growing segment of community sport (the six player change rooms proposed here serve exactly that demand), all-abilities inclusion through accessible amenities and hearing augmentation, and a consultation model in which a steering group of the residents association and the clubs shaped the design principles before a design brief was written. Their new carpark even double-serves the local primary school at drop-off — the same shared-use thinking proposed here for the tennis-entrance car park.

What Montrose Cricket Club gets

Two proper grounds instead of compromised ones: a 110 m boundary on the south ground and 100 m on the north, both protected in the design — the football geometry was made to fit around the cricket, not the other way round. The wickets are concrete with carpet, sited between the football pitches so neither code damages the other, and covered by the mobile shelters through winter. The grounds are continuous flat grass inside their fences with no internal obstructions. In a small number of fence sections the rope-to-fence gap runs under the guideline 2.74 m (tightest at the south ground's road-side taper and the north ground's corner cut); the plan documents these runs and the standard fixes — removable fence panels for the cricket season, or the usual local rule scoring ball-to-fence as a boundary — and the north-ground pinch is expected to relax once the channel survey confirms the arm's position.

Pitch lighting

The redevelopment includes new sports lighting across all four pitches, specified by role: the two main pitches (Pitch 2 and Pitch 3) aim for a **200 lux** average — the level required for senior club competition at night — while the two outer pitches (Pitch 1 and Pitch 4) aim for **100 lux**, the recognised standard for match practice and junior competition. This mirrors the tiered approach at Aspendale Gardens, where twelve modern lighting poles service the four-pitch layout with a 100 lux minimum across AFL (Australian football) and football requirements, and it concentrates the energy and capital cost where night competition will actually be played. Pole positions will be developed with the levels model so that no pole conflicts with the cricket outfield, the fence lines, or the watercourse offsets, and lighting spill toward the Swansea Rd dwellings and the northern bushland will be a design criterion from the first layout.

Concrete out of the field of play

There is a second safety dimension to the cricket arrangement, and recent events have made it impossible to treat as theoretical. Like most shared grounds in Victoria, the reserve's existing ovals carry their concrete cricket wickets in the middle of the playing surface, under seasonal synthetic covers through the football season. Over the years this club has had its share of hard falls onto that concrete — never a documented concussion, but several, and a few broken ankles among them. In July 2026 the risk this represents was realised in the worst possible way at Lalor Recreation Reserve, where a 27-year-old footballer died after striking his head on the covered concrete wicket in the middle of the ground; WorkSafe is investigating, and councils across Victoria are now reviewing every covered in-field wicket they manage.

This master plan removes the hazard rather than managing it. The concrete wicket slabs are sited **in the corridors between the football pitches — never within a pitch** — finishing a minimum of 1 m outside the pitch side lines, with the mobile team shelters standing over them through winter. No football is played on, across, or into concrete anywhere in the layout. For a council reviewing covered wickets across its municipality, this reserve would move from the at-risk list to the exemplar list — and it is a design decision that costs nothing, because the shared-ground geometry was built around it from the start.

Cricket wicket orientation — measured, assessed, and improved

Wicket orientation was checked against the published guidance and against the existing ground as actually built — the two existing wickets were surveyed from field-measured end points rather than estimated. Cricket Australia’s playing-field guidance calls for an approximately north–south wicket, within a window from 45° east of north to 35° west of north and an optimum of 10–15° east of north. The measurements tell a split story about the existing reserve: the **southern wicket runs at 29.6°, inside the guidance window**, but the **northern wicket runs at 65.5°** — well outside it, and oriented so that its south-western-end batter faces 245.5°, essentially dead in line with the Melbourne midsummer setting sun (azimuth ~240–255° late in the day). That is the precise failure mode that halted an international at Napier’s McLean Park, where a non-standard wicket alignment put the setting sun in the batters’ eyes and play was suspended on safety grounds.

This matters because cricket at the reserve is a summer, daylight game — it does not use the sports lighting, which serves the winter football season — and senior fixtures start at 1pm and continue to around 7pm. In a Melbourne summer that takes the final hour of play directly into a low western sun, at only ~15–20° elevation, with no lights to fall back on. Wicket orientation is not a theoretical consideration here; it is the last hour of every senior game.

The proposed layout responds on both grounds — and as of rev 9.30, **both wickets sit inside the guidance window**. The shared-ground geometry sets the precinct on a 310° bearing, but the wicket slabs need not follow it exactly. On the northern ground, Pitch 4’s width was trimmed from 60 m to 56.4 m to widen the pitch 3/4 gap, rotating the C2 wicket slab to **326°**. On the southern ground, Pitch 1 has been shifted a total of about 3.2 m south-west to widen the pitch 1/2 gap to 12.15 m, rotating the C1 wicket slab to **325°** — with the concrete ends of both slabs finishing exactly 1 m clear of every football pitch side line, and the mobile shelters rotating with them. The Pitch 1 shift is Stage 3 work on the southern ground with no effect on Stages 1 and 2, and carries one disclosed dependency: it takes the south-west fence-to-drain clearance to 1.9 m, marginally under the 2 m held everywhere else on the plan, to be confirmed by the detailed site survey. Under the solar assessment the two wickets keep the low evening sun 60–86° off every batter’s sight line — well outside the glare cone — with the residual nuisance transferred to square fielders and the low western sun path screened by the retained treed drain corridor. In short: measured against the ground the clubs actually play on today, the new layout eliminates the reserve’s genuinely dangerous wicket (65.5°, batter in the setting sun) and replaces it with two fully compliant wickets — a measurable safety improvement, offered up-front rather than left for review to find. #### A district benchmark: six measured wickets, and what the sun actually does here

To test the orientation reasoning against reality rather than guidance alone, the wickets of six grounds across the district were measured by coordinate — four corner points per carpet, with the two long edges of every carpet agreeing in bearing to within 0.4°. The physics being tested is specific to this latitude and this fixture pattern. In Melbourne’s cricket season the sun sets south of west: about azimuth 240° at midsummer, drifting to ~255–265° in the shoulder months, and glare becomes hazardous when that low sun sits within roughly 30° of a batter’s sight line. Because fixtures here run 1pm to about 7pm, only the evening sun matters — and working the

geometry for that window puts the locally optimal wicket axis at roughly 335–340°, west of north. The national guidance window (45° east of north to 35° west) is a whole-of-Australia compromise that must also protect morning play; for an afternoon-only ground at this latitude, the west side of north is simply the correct side to deviate toward. The measurements bear this out:

Ground	Council	Bearing	Evening sun off batter's sight line
KHFR existing north wicket	Yarra Ranges	65.5°	0–20° — directly in the glare
Lilydale Recreation Reserve	Yarra Ranges	35.0°	25–50°
Mt Evelyn Recreation Reserve	Yarra Ranges	31.6°	28–54°
KHFR existing south wicket	Yarra Ranges	29.6°	30–55°
Montrose Recreation Reserve	Yarra Ranges	26.2°	34–59°
KHFR proposed C1	<i>this plan</i>	325°	60–85°
KHFR proposed C2	<i>this plan</i>	326°	61–86°
Glen Park Reserve	Maroondah	348.8°	71–96°
Wandin North Recreation Reserve	Yarra Ranges	340.9°	76–104° — best measured

Two findings stand out. First, the district's east-of-north convention — every measured Yarra Ranges wicket except one sits between 26° and 35° east — clusters those grounds between marginal and, at KHFR's north wicket, dangerous for evening play, with performance degrading as bearings drift further east. Second, the two west-of-north wickets in the sample are the two best performers — and one of them, Wandin North at 340.9°, is Yarra Ranges' own. The proposal is therefore not asking the council to accept anything novel: it brings the orientation standard already proven at the council's best-performing ground to a reserve whose current north wicket is the worst in the sample, and the two proposed wickets would hold the evening sun further from batters' eyes than any east-of-north ground currently played on in the district.

Playing surface, climate resilience and flood behaviour

The plan commits to **natural turf on all four pitches**. Synthetic surfaces were considered and set aside on health, environmental and regulatory grounds: the rubber-crumb microplastic infill they shed is the subject of an adopted European Union ban (sales phased out under the 2023 REACH microplastics restriction), synthetic carpets run substantially hotter than grass in summer, and end-of-life disposal is an unresolved waste problem. A natural-turf precinct also behaves fundamentally better in flood: the grounds can absorb and temporarily store stormwater in a way plastic carpet on a compacted base cannot — in a flash-flood event the reserve works with the drainage system rather than against it, helping keep Swansea Rd open, while the pavilion on its 3 m podium stands clear of flood level.

The turf specification itself is left as an open design decision, informed by hard local experience. The reserve's top pitch currently runs Santa Ana couch, and the club's lived result is consistent complaints: excellent in summer, but winter-dormant to the point of looking dead, and unable to recover from simple scuffs in the middle of the football season — the exact months of heaviest use. Professional venues solve winter wear with perennial ryegrass, which repairs quickly in cold weather but is thirsty and maintenance-intensive; modern sterile couches such as Tif Tuf sit between, and their winter performance is improving with treatment products. Under current Victorian climate projections — hotter, drier summers with more intense short-duration rainfall — the sensible position is a surface strategy set at detailed design (likely a couch base with winter ryegrass oversowing on the main football pitches), sized against the water harvesting described below. Climate-resilient community infrastructure of precisely this kind — flood-tolerant grounds, drought-conscious surfaces, buildings above flood level — is the class of project current federal adaptation and resilience funding programs exist to support, and the project summary flags this as a live co-funding avenue alongside the state sport partnership.

Water, energy and storage — using the building twice

The pavilion earns its keep beyond match day through three further provisions — one a requirement of the design, two optional additions that are cheapest if provided for now rather than retrofitted:

Rainwater harvesting. The enclosed pavilion roof is ~1,750 m². At Montrose's rainfall that roof yields on the order of 1.3–1.4 million litres a year — collected into the reserve's existing tank as part of the base project — the tank itself needs a slight relocation of about 38 m north-north-east (from MGA55 355198, 5814766 to 355209, 5814802), moving it out of the new southern playing field and into the pocket beside the pavilion deck, conveniently adjacent to the roof it will harvest from, with an **optional** second large tank in the podium undercroft beneath the pavilion, where the 3 m build-up creates storage volume that costs almost nothing to enclose. Harvested roof water directly answers the biggest objection to the better winter playing surfaces: ryegrass's thirst is far more affordable when a season's irrigation for the high-wear zones and wicket surrounds is falling on your own roof.

Rooftop solar (optional). The same roof supports a large photovoltaic array (a 100 kW-class system fits comfortably), offsetting the pavilion's daytime loads with headroom for future battery storage. Federal programs co-funding energy upgrades at council community and sporting facilities make this the single most fundable line item in the project.

Equipment storage in the undercroft — required, not optional. The club already struggles every summer to store its portable goals — a fleet of 5 × 2 m junior and 3 × 2 m Mini Roos goals that grows as junior numbers do. The deck and pavilion podium create undercroft space at pitch level; a secure roller-door storage bay under the deck puts the goals metres from where they're used, off the grounds over summer, and out of the cricket clubs' way. This is a requirement of the pavilion design rather than an extra, and it costs structure that the podium provides anyway.

Public transport, active travel and equal access

The reserve is already on a genuinely useful bus route — the **679, running seven days a week** — but today it serves the ground in one direction only (towards Ringwood), from a single sheltered stop on the east side of Swansea Rd, positioned between the two turning lanes and fronting the existing shared path. The plan proposes completing that picture with three modest, connected works:

A new outbound bus stop close to opposite the existing inbound stop, so the route serves the reserve in both directions. Arriving players and spectators would alight on the reserve side with no road to cross — the busiest movement made the safest.

An improved pedestrian crossing — no traffic lights. A dedicated pedestrian refuge island in the middle of Swansea Rd, built wide enough to stage a bicycle, linking the east-side shared path and inbound stop to the reserve and the new pavilion. For cyclists coming off the shared path, the wide island is the difference between crossing and not crossing: it lets a rider stage the crossing in two halves, judging one direction of traffic at a time with the bike fully protected in the median, instead of waiting for a simultaneous gap in both directions of an arterial road. A refuge island is a standard, low-cost Department of Transport and Planning treatment appropriate to the road environment; its exact position will be set at detailed design to work with the existing turning lanes, landing on the pavilion's desire line.

Equal access, designed in from the start. The crossing connects to a step-free accessible path running to the pavilion, with **ramp access up onto the pavilion podium** at compliant grades so the change rooms, canteen, deck and viewing areas are available to everyone — players, officials, and spectators alike. Part of the existing central car park at the second Swansea Rd entrance becomes a **dedicated accessible parking section**, the closest parking on the reserve to the pavilion, with the accessible path linking it to the same ramp. A junior who catches the 679 to training, a spectator with a pram, and a wheelchair user driving to the ground all arrive at the same front door.

Together these turn a car-dependent reserve into a genuinely multi-modal one — every bus trip and bike trip is a car movement removed from Swansea Rd at peak sport times — and access-for-all provisions of this kind are scored directly in state and federal community-sport funding criteria.

Why this is an easy approval for Council

The scheme doubles the reserve's playing capacity while shrinking its impact. The steep north corner remains untouched bushland reserve. Every goal-line fence sits at the 4 m minimum specifically to pull the fenced footprint off the treed Swansea Rd frontage — roughly 860 m² less clearing than earlier iterations of this same plan. Both existing Swansea Rd entrances with their turning lanes are retained as-is. The watercourse is barely touched (detailed below), existing community infrastructure like the reverse vending machine stays where it is, and the whole project can be delivered in two stages that keep sport running throughout.

Detail 1 — The old tennis club entrance, driveway and a ~220-space car park

The former Montrose Terrace tennis club entrance enters the reserve at its south corner directly off Swansea Rd, and the proposal builds the site's main car park around it — sized at approximately **6,200 m² for about 220 car spaces** (at the standard planning rate of 28 m² per space including aisles, AS 2890), comfortably above the 120–180 spaces a four-pitch, two-oval venue typically requires at peak, with room held in reserve to the north if a traffic assessment ever asks for more.

The arrangement deliberately spends nothing on new road or drainage infrastructure. Vehicle access reuses the **existing formed crossover and its established turning-lane treatments** on Swansea Rd — no new intersection, no new turn-lane works, and no additional assessment burden beyond the existing entrance's capacity. Where the access driveway crosses the lower Montrose drain just inside the entrance, it does so at the **existing culvert crossing**, retained in place — the only work anticipated there is a condition and capacity check on the existing structure, which folds neatly into the Melbourne Water conversation already required for the channel deviation upstream. From the culvert, only about **75 m of new driveway** is needed to reach the car park, because the parking has been drawn as far south on the old tennis terrace as the site constraints allow.

The car park's position also answers the nearest neighbour before they ask: it stands roughly **12 m off the western boundary** adjoining the residential property, with that strip retained as vegetation, and its whole footprint sits on the already-disturbed former tennis ground rather than in native vegetation. Match-day traffic therefore enters at the corner of the site, parks within 100 m of the gate, and never crosses the pedestrian routes between the pitches — while the two existing Swansea Rd entrances with their turning lanes continue serving the central and pavilion areas unchanged, giving the reserve three functioning access points without a single new road connection.

Detail 2 — The watercourse is changed as little as possible

The reserve's defining constraint is the water path through its middle, and the design's answer is restraint. The **main watercourse — the west branch and the lower leg to the southern exit — is not moved at all**. The lower Montrose drain along the south-west, whose true position was field-verified during design (it runs naturally further west than older mapping suggested), is likewise **retained exactly on its existing course** — the verification removed any need to touch it. The only physical channel work in the entire scheme is a single **~105 m realignment of the upper Montrose drain reach**, sweeping it around the north corners of Pitch 2 at a constant 5 m clear of the pitch. It is drawn as a naturalised channel — gentle meanders, no straight runs, no sharp bends — replacing what is today an awkward right-angle jog, so the works arguably leave the channel in better form than they find it. This is the one item requiring Melbourne Water consent, and it is deliberately modest: same function, same connectivity, ~105 m of improved channel. The north-east arm is not altered as a channel at all; the plan shows it at its design position pending a confirming survey, and the fences and pitches hold the standard 5 m offset

from it throughout.

Detail 3 — The reverse vending machine stays where it is

The reverse vending machine in the central car park off Swansea Rd is retained **in its current position, untouched**. The playing precincts, fences, pavilion podium and channel works were all laid out around it as a fixed point — its location is carried in the coordinate model alongside the pitches and fences, and no element of the plan encroaches on it or its servicing access from the central car park. Council's container-deposit infrastructure therefore continues operating through both construction stages without relocation cost or service interruption — a small point, but one that signals the design philosophy of the whole scheme: keep everything that works, and spend the project's budget only on what doesn't.

Staged delivery and indicative costs

The Aspendale benchmark makes an honest first-pass cost frame possible before any quantity surveying: Kingston's full ground transformation ran to roughly \$9.1 million (\$4.6M council, \$4.5M Victorian Government), of which the 1,064 m² pavilion was \$3.6 million. Scaling those rates to the Montrose scope, and stating plainly that these are Aspendale-benchmarked planning figures — pre-survey, pre-QS, and subject to construction escalation since that project — the staging breaks down approximately as follows.

Stage	Scope	Indicative cost (Aspendale-benchmarked)
1	Pitches 3 & 4, north cricket ground (C2), wicket, fencing, drainage platform, north lighting (200/100 lux)	\$2.5M – \$3.0M
2	Pavilion on 3 m podium: 70 × 25 m enclosed + 20 × 20 m elevated viewing deck + rooftop viewing area (modular design-and-construct)	\$3.5M – \$6.5M (see note)
3	Pitches 1 & 2, south cricket ground (C1), wicket, fencing, south lighting, ~220-space car park, 75 m driveway, drain deviation	\$3.0M – \$3.5M
	Indicative total	\$9M – \$13M

The Stage 2 band deserves a straight explanation: even at 70 × 25 m the enclosed building (~1,750 m²) is well above the floor area of Aspendale's \$3.6M, 1,064 m² pavilion, and at Aspendale's ~\$3,400/m² rate the enclosed works alone approach \$6M. The recomposition of the northern 20 m into an open elevated deck is the design's own value-management move —

roofless deck structure prices at a fraction of enclosed, serviced floor area, trimming several hundred thousand dollars from the brief while adding the venue's best social asset: outdoor table viewing across both main pitches, with the rooftop area above sighting all four. Settling the final envelope belongs to the Stage 2 design phase, informed by what state co-funding is available. On the funding side, the Aspendale split is the model: with a comparable state partnership, council's share of even the full program lands in the \$5–7M range spread across three budget years, each stage independently useful if later stages are deferred.

Status and next steps

The geometry is complete and internally verified at rev 9.30 (drawing set KHFR rev 9-30: master plan, aerial overlay, and MGA55 coordinate schedule): every dimension, fence vertex, boundary circle and channel line is issued as MGA55 coordinates ready for a surveyor. Three items close out the design: a field survey of the north-east arm's bank position (which gates Pitch 3/4 lengths and improves the C2 fence margins), the Vicmap contour data for the levels and earthworks model of the two ground platforms and pavilion podium, and a Melbourne Water pre-application conversation on the ~105 m drain deviation. Ball-stop treatment along the Swansea Rd frontage is anticipated as a likely council condition and is accommodated by the fence design.